14.23 Solution: (a) From Problem 14.13, we have

$$\frac{dP_m(N)}{d\Omega} = \frac{\omega_0^4 m^2}{(2\pi c)^3}\left|\sum_{j=1}^N q_j \int_0^{2\pi/\omega_0} \mathbf{v}_j(t)\times\mathbf{n}\exp\left[im\omega_0\left(t-\frac{\mathbf{n}\cdot\mathbf{x}_j(t)}{c}\right)\right]dt\right|^2.$$

For the particles, their locations and velocities are

$$\mathbf{x}_j(t) = R(\cos(\omega_0 t+\phi_j), \sin(\omega_0 t+\phi_j), 0), \quad \mathbf{v}_j(t) = \omega_0 R(-\sin(\omega_0 t+\phi_j), \cos(\omega_0 t+\phi_j), 0).$$

Use the result from Problem 14.15 (a), we now have

$$\begin{aligned}\frac{dP_m(N)}{d\Omega} &= \frac{\omega_0^4 m^2}{(2\pi c)^3}\left|\sum_{j=1}^N q_j \int_0^{2\pi/\omega_0} \begin{pmatrix}\cos\theta\cos(\omega_0 t+\phi_j)\\ \cos\theta\sin(\omega_0 t+\phi_j)\\ -\sin\theta\cos(\omega_0 t+\phi_j)\end{pmatrix}\omega_0 R e^{im\omega_0 t}\right.\\ &\qquad\left.\times\sum_{k=-\infty}^{\infty}(-i)^k J_k(m\beta\sin\theta)e^{-ik(\omega_0 t+\phi_j)}dt\right|^2.\end{aligned}$$

It is straightforward to verify that

$$\begin{aligned}&\int_0^{2\pi/\omega_0}\begin{pmatrix}\cos(\omega_0 t+\phi_j)\\ \sin(\omega_0 t+\phi_j)\end{pmatrix}e^{im\omega_0 t}\sum_{k=-\infty}^{\infty}(-i)^k J_k(m\beta\sin\theta)e^{-ik(\omega_0 t+\phi_j)}dt\\ =\quad & e^{-im\phi_j}\int_0^{2\pi/\omega_0}\begin{pmatrix}\cos(\omega_0 t)\\ \sin(\omega_0 t)\end{pmatrix}e^{im\omega_0 t}\sum_{k=-\infty}^{\infty}(-i)^k J_k(m\beta\sin\theta)e^{-ik\omega_0 t}dt.\end{aligned}$$

Then,

$$\begin{aligned}\frac{dP_m(N)}{d\Omega} &= \frac{\omega_0^4 m^2}{(2\pi c)^3}\left|\sum_{j=1}^N q_j e^{-im\phi_j} \int_0^{2\pi/\omega_0} \begin{pmatrix}\cos\theta\cos(\omega_0 t)\\ \cos\theta\sin(\omega_0 t)\\ -\sin\theta\cos(\omega_0 t)\end{pmatrix}\omega_0 R e^{im\omega_0 t}\right.\\ &\qquad\left.\times\sum_{k=-\infty}^{\infty}(-i)^k J_k(m\beta\sin\theta)e^{-ik\omega_0 t+\phi_j}dt\right|^2,\\ &= \frac{dP_m(1)}{d\Omega}\cdot F_m(N),\end{aligned}$$

where

$$F_m(N) = \left|\sum_{j=1}^N q_j e^{-im\phi_j}\right|^2.$$

(b) If all q_j's are the same, $q_j \equiv q$, and the particles are evenly space with $\phi_j = 2\pi j/N$, then

$$F_m(N) = q^2\left|\sum_{j=1}^N \exp\left(-im\cdot\frac{2\pi j}{N}\right)\right|^2.$$

It is easy to verify that only when m is a multiple of N, and the sum if N,

$$\sum_{j=1}^N \exp\left(-im\cdot\frac{2\pi j}{N}\right) = N, \quad \text{if } m = nN \text{ for } n\in\mathbb{Z}.$$

Therefore, the radiation frequency must be a multiple of $N\omega_0$, and the intensity is N^2 times of the one particle result.

(c) Since the radiation frequency must be a multiple of $N\omega_0$, the lowest m in the expression of Problem 14.15 (a) must be N. For nonrelativistic motion, $\beta \ll 1$, $J_N(x) \sim x^N$ and $dJ_N(x)/dx \sim x^{N-1}$, both terms in the braces are of the order β^{2N-2}. Taking into account of the pre-factor, we can see that the leading order contribution is of the order β^{2N}. Therefore, the radiation tends to 0 as $N \to \infty$.

(d) From Landau and Lifshitz, *The Classical Theory of Fields*, Section 70, Eq. (70.9), for large N,

$$J_N(N\epsilon) \approx \frac{1}{\sqrt{\pi}}\left(\frac{2}{N}\right)^{1/3} \Phi\left[\left(\frac{N}{2}\right)^{2/3}(1-\epsilon^2)\right],$$

where $\Phi(x)$ is the Airy function, defined as

$$\Phi(t) = \frac{1}{\sqrt{\pi}}\int_0^{+\infty} \cos\left(\frac{x^3}{3} + xt\right) dx,$$

and is related to the modified Bessel function of second kind with order 1/3 by

$$\Phi(t) = \sqrt{\frac{t}{3\pi}} K_{1/3}\left(\frac{2}{3}t^{3/2}\right).$$

We now can approximate the result of Problem 14.15 (a) as

$$J_N(N\beta\sin\theta) \propto \Phi\left[\left(\frac{N}{2}\right)^{2/3}(1-\beta^2\sin^2\theta)\right] \propto K_{1/3}\left(\frac{N}{3}(1-\beta^2)^{3/2}\right) = K_{1/3}\left(\frac{N}{3\gamma^3}\right).$$

The modified Bessel function of second kind has the following asymptotic expansion,

$$K_\nu(t) \sim e^{-t},$$

and

$$J_N(N\beta\sin\theta) \sim \exp\left(-\frac{N}{3\gamma^3}\right).$$

For the other term, $J'_N(x)$, we can approximate it as $K'_{1/3}(x)$, which can be expressed as $-(K_{2/3}(x) + K_{4/3}(x))/2$ and also has the same asymptotic behavior. Therefore, the power radiated behaves asymptoticcally as $e^{-2N/3\gamma^3}$, which vanishes in the limit of $N \to \infty$, even when the particles are moving relativistically.

(e) From parts (c) and (d), it is clear that when the particle number tends to infinity and their positions are symmetrically distribued on a circular loop, there will be no radiation. The configuration is closely related to that of a steady current, which according to our study of the static property of magnetic field, should have no radiation at all.